# Chapter 6 Notes

## Section 1

### Notes

Note that $F \upharpoonright A$ does not necessarily equal $A \times F''(A)$. In fact, if $F''(A)$ contains more than one element (say it contains at least the two elements $b, c$), then obviously $F \upharpoonright A \neq A \times F''(A)$, as, for any element $x \in A$, we would have $\langle x, b\rangle$ and $\langle x, c\rangle$ both in $A \times F''(A)$, but there can only be one $y$ for which $\langle x, y\rangle \in F \upharpoonright A$.

It is remarked that:"If $\phi_1$ and $\phi_2$ are functions from a class $A$ onto a class $B$ and if $\phi_1\phi_2^{-1}$ is an identity function, then obviously $\phi_1 = \phi_2$." Here is why this is. Now, note that this assumes that $\phi_2^{-1}$, is, in fact, a function. Then, for every element $y$ in the range of $\phi_2$, we have $\phi_2^{-1}(y) = x$ for some $x$ in the domain of $\phi_2$, and $\phi_1\phi_2^{-1}(y) = \phi_1(x) = y$. But also, $\phi_2(x) = y$. Thus $\phi_1(x) = \phi_2(x)$ for every $x$ in the domain of $\phi_2$, and because they have the same domains, the functions are the same.

Note that if a class $A$ is well ordered under $\subseteq$, then it is also properly well ordered under $\subseteq$. The proof of this is the same as the proof of Proposition 1.3.

### Exercise 1.1

Let $\mathcal{N}$ be a nest of functions.

(1): We first show that $\cup\mathcal{N}$ is function. Obviously $\cup\mathcal{N}$ is a relation. Now, suppose we had $\langle x, y\rangle$ and $\langle x, z\rangle$ both in $\cup\mathcal{N}$. Thus, for some functions $f$ and $g$, $\langle x, y\rangle \in f$ and $\langle x, z\rangle \in g$. But, because $\mathcal{N}$ is a nest, either $f \subseteq g$ or $g \subseteq f$. Without loss of generality, suppose $f \subseteq g$. Then we would have $g(x) = y$ and $g(x) = z$, and because $g$ is a function, $y = z$. Hence $\cup\mathcal{N}$ must be a function.

(2): We now show that the domain of $\cup\mathcal{N}$ is the union of the class of domains of the elements of $\mathcal{N}$ (call this $D$). Suppose $x \in \mathrm{Dom}(\cup\mathcal{N})$. Then, for some $y$ and some function $f \in \mathcal{N}$, $\langle x, y\rangle \in f$, meaning $x \in D$. On the other hand, suppose $x \in D$. Then, for some $y$ and some function $f \in \mathcal{N}$, $\langle x, y\rangle \in f$, meaning $x \in \mathrm{Dom}(\cup\mathcal{N})$. Thus $\mathrm{Dom}(\cup\mathcal{N}) = D$. Similar reasoning shows $\mathrm{Ran}(\cup\mathcal{N})$ is the union of the class of ranges of the elements of $\mathcal{N}$.

(3): Lastly, suppose that every element of $\mathcal{N}$ was 1-1. Now, suppose $x$ and $y$ are in the domain of $\cup\mathcal{N}$, and that $x \neq y$, and for some $v$ and $w$, $\langle x, v\rangle \in \cup\mathcal{N}$ and $\langle y, w\rangle \in \cup\mathcal{N}$. Then, for some functions $f$ and $g$ in $\mathcal{N}$, $\langle x, v\rangle \in f$ and $\langle y, w\rangle \in g$. Without loss of generality, suppose that $f \subseteq g$. Then $g$ is 1-1, meaning $v \neq w$. Hence $\cup\mathcal{N}$ is 1-1.

### Exercise 1.2

First, let us note that $On$ is not a set. For if it were, it would then be a transitive set of ordinals, and thus would be an ordinal. Then that would mean $On \in On$, which never holds for ordinals. Thus $On$ is not a set.

Now, let $A$ be the class $On \cup \{\{\{\emptyset\}\}\}$. Thus $x \in A$ iff either $x$ is an ordinal, or $x = \{\{\emptyset\}\}$. Note that $\{\{\emptyset\}\}$ is not an ordinal, as if it were, then it would be transitive, which would imply that $\emptyset \in \{\{\emptyset\}\}$, which is not the case. Hence $On$ is a proper subclass of $A$.

Then define the relation $x \leq y$ (on $A$) as follows: $x \leq y$ iff $x$ and $y$ are both ordinals and $x \subseteq y$, or $y = \{\{\emptyset\}\}$. Note that this defintion implies that $\{\{\emptyset\}\} \leq y$ only holds if $y = \{\{\emptyset\}\}$. We first show that $\leq$ is a well ordering of $A$.

$O_1$: If $x$ is an ordinal, then $x \subseteq x$, so that $x \leq x$ holds. If $x = \{\{\emptyset\}\}$, then $x \leq x$ by definition.

$O_2$: Suppose $(x \leq y) \wedge (y \leq z)$. If all of $x, y$ and $z$ are ordinals, then obviously $x \leq z$. If any of $x, y$ and $z$ were equal to $\{\{\emptyset\}\}$, then $z = \{\{\emptyset\}\}$ must hold, which of course implies that $x \leq z$.

$O_3$: Suppose $x \leq y$ and $y \leq x$. Then either $x = y = \{\{\emptyset\}\}$, or both $x$ and $y$ are ordinals, in which case $x \subseteq y$ and $y \subseteq x$, so that $x = y$. Thus $x = y$.

$O_4$: If $x$ and $y$ are both ordinals, then of course either $x \leq y$ or $y \leq x$. If $y = \{\{\emptyset\}\}$, then $x \leq y$ (similarly with $x$).

$O_5$: For any subclass $B$ of $A$ that only contains ordinals, then, because $On$ is well ordered under $\subseteq$, it contains a least element. If $\{\{\emptyset\}\} \in B$, then $B - \{\{\{\emptyset\}\}\}$ contains a least element $e$, and $e$ is also the least element of $B$ itself, as $e < \{\{\emptyset\}\}$. Thus every subclass $B$ of $A$ contains a least element.

Thus $\leq$ is a well ordering of $A$. Now, $On$ is a proper lower section of $A$: It is a proper subclass of $A$, and $x < \{\{\emptyset\}\}$ for all elements in $On$. But $On$ is not a set. Thus we have an example of a well ordering which is not a proper well ordering.

# Section 2

## Notes

Note that any subclass of an isomorphism from $A$ onto $B$ is an isomorphism from a subclass of $A$ onto a subclass of $B$.

Obviously, if $A$ is isomorphic to $B$, then $B$ is isomorphic to $A$ (as seen in the next paragraph).

Note that, if $\phi$ is an isomorphism from $A$ onto $B$, then $\phi^{-1}$ is an isomorphism from $B$ onto $A$. For, clearly $\phi^{-1}$ is a 1-1 function from $B$ onto $A$. And suppose $\phi$ is an isomorphism from the system $\langle A, \leq_1 \rangle$ to the system $\langle B, \leq_2 \rangle$. Then, for any $a_1, a_2$ in $B$, $a_1 <_2 a_2$ iff $\phi(a_1) <_1 \phi(a_2)$.

The above fact is used, for instance, in Corollary 2.2. Also, in Corollary 2.2, note that $(2')$ follows because $\phi(\phi^{-1}(a)) = a$.

I think there is an error in the proof of Corollary 2.3. It claims that $\phi_1\phi_2^{-1}$ is an isomorphism from $A$ onto $A$, but this is clearly not the case. For $\phi_2^{-1}$ is from $B$ onto $A$, and $\phi_1$ is from $A$ onto $B$, meaning $\phi_1\phi_2^{-1}$ is actually from $B$ onto $B$ (now, this would also work, as then, by Corollary 2.2, $\phi_1 = \phi_2$, but it is explicitly said in the proof that $\phi_1\phi_2^{-1}$ is from $A$ onto $A$). If we wanted to get a function from $A$ onto $A$, we would need to pick $\phi_2^{-1}\phi_1$ (for example; there are others).

Here is why $\phi_1\phi_2^{-1}$ is in fact an isomorphism. For, if $a_1$ precedes $a_2$ in $B$, then $\phi_2^{-1}(a_1)$ precedes $\phi_2^{-1}(a_2)$ in $A$, and $\phi_1(\phi_2^{-1}(a_1))$ precedes $\phi_1(\phi_2^{-1}(a_2))$ in $B$ (more generally, if $f$ is an isomorphism from $A$ onto $B$, and $g$ is an isomorphism from $B$ onto $A$, then $fg$ is an isomorphism from $B$ onto $B$).

Also, here is why (as said in Corollary 2.3), if $\phi_1\phi_2^{-1}$ is the identity map (for $\phi_1$ and $\phi_2$ both from $A$ onto $B$), then $\phi_1 = \phi_2$. For suppose that $\phi_1\phi_2^{-1}$ is the identity map. Then, for any $x \in B$, $\phi_2^{-1}(x) = y$ for some $y \in A$, and $\phi_1(y) = x$, but also $\phi_2(y) = x$, meaning $\phi_1 = \phi_2$.

Corollary 2.5 holds, as, for any two distinct ordinals $\alpha, \beta$, they are comparable. Without loss of generality, suppose $\alpha < \beta$. Now, $\alpha$ must be an initial segment of $\beta$ (clearly, all elements of $\alpha$ are in $\beta$, due to the transitivity of ordinals). This is because, for any element $x \in \alpha$, and for any element $y \in \beta$ but not in $\alpha$, $x < y$ holds because $x < \alpha$ (by assumption), and, if $x \geq y$, then $y < \alpha$. But, this would mean $y \in \alpha$, a contradiction. Hence $\alpha$ is an initial segment of $\beta$, and thus there can be no isomorphism from $\beta$ to $\alpha$.

Corollary 2.6 holds, as, between two distinct lower sections of the same well ordered system, one is an initial segment of the other. For supppose the two lower sections were $L_<(x)$ and $L_<(y)$. Now, without loss of generality, suppose $x < y$

(and thus $L_<(x)$ is a subset of $L_<(y)$). Then $L_<(x)$ is an initial segment of $L_<(y)$, as, for every $z \in L_<(x)$, obviously $z < w$ for every $w \in L_<(y)$ but not in $L_<(x)$ (for if not, then $z \geq x$), meaning $L_<(x)$ is proper lower section of $L_<(y)$, and hence also an initial segment of $L_<(y)$. Thus, if two distinct lower sections of the same well ordered system are isomorphic to each other, then one of them would be isomorphic to one of its initial segments, contrary to Corollary 2.4.

If $A$ is isomorphic to $B$, and $B$ isomorphic to $C$, then $A$ is obviously isomorphic to $C$. For suppose $A$ is isomorphic to $B$ under $\phi_1$, and $B$ is isomorphic to $C$ under $\phi_2$. Then let $\phi_3$ be the class of all orderd pairs $\langle x, \phi_2(\phi_1(x))\rangle$ where $x \in A$; $\phi_3$ is the isomorphism from $A$ to $C$.

This explains Corollary 2.7: For suppose a lower section of $A$ was isomorphic to two distinct lower sections $C$ and $D$ of $B$. Then it would follow that $A$ is isomorphic to $C$, and $A$ is isomorphic to $D$, meaning $C$ is isomorphic to $D$. But this would contradict Corollary 2.6.

In Corollary 2.8, here is why $I_2 \upharpoonright D_1$ is an isomorphism from $D_1$ onto a lower section of $B$: Clearly $I_2 \upharpoonright D_1$ is an isomorphism from $D_1$ onto a subclass of the range of $I_2$. Also, suppose that $A$ was well ordered under $\leq_1$ and $B$ was well ordered under $\leq_2$. Now, suppose $C$ is the range of $I_2$, and $C'$ is the range of $I_2 \upharpoonright D_1$. Suppose that $C'$ was not a lower section of $C$. Then, for some $x \in C'$, there would be some $y \in C - C'$ such that $y <_2 x$. But, because $I_2$ is an isomorphism, $I_2^{-1}(y) <_1 I_2^{-1}(x)$. But $I_2^{-1}(x) \in D_1$, and it was assumed that $D_1 \subseteq D_2$, meaning $I_2^{-1}(x) <_1 I_2^{-1}(y)$ (because $D_1$ and $D_2$ are both lower sections of $A$), meaning it cannot be that $y <_2 x$, and hence $I_2 \upharpoonright D_1$ must be an isomorphism onto $C'$, a lower section of $B$ (that is, a lower section of $C$, but because $C$ is a lower section of $B$, then $C'$ is also a lower section of $B$).

Here is why $\mathcal{I}$ in the proof of Theorem 2.9 must be an isomorphism from a lower section of $A$ onto a lower section of $B$. Recall that $\mathcal{I} = \cup\mathcal{N}$, and $\mathcal{N}$ is the class of all isomorphisms from lower sections of $A$ which are sets onto lower sections of $B$ which are sets. By Proposition 1.4, $\mathcal{I}$ is a function. The domain of $\mathcal{I}$ must be a lower section of $A$. For suppose it weren't. Then for some $x$ in the domain of $\mathcal{I}$, there would be some $y \in A$ not in the domain of $\mathcal{I}$ such that $x > y$. But $x$ is also in the domain of some $f \in \mathcal{N}$ which is an isomorphism from a lower section of $A$, which would mean that $y$ is in the domain of $f$ (and thus also of $\cup\mathcal{N}$), a contradiction. Hence the domain of $\mathcal{I}$ is a lower section of $A$. Similarly, it can be seen that $\mathcal{I}$ is onto a lower section of $B$.

# Section 3

## Exercise 3.1

(a): On the one hand, suppose the axiom of substitution. Now, suppose $F$ is a function and $x$ is a set. Then $F \upharpoonright x$ is a subclass of $x \times F''(x)$. But $F''(x)$ is a set by the axiom of substitution, meaning $x \times F''(x)$ is a set, and hence so is $F \upharpoonright x$. On the other hand, suppose that for any function $F$ and any set $x$, the function $F \upharpoonright x$ is a set. Then consider any function $F$ and any set $x$. Note that $F''(x)$ is a subclass of $\cup(F \upharpoonright x)$. Thus, because $F \upharpoonright x$ is a set, so is $\cup(F \upharpoonright x)$, and hence so is $F''(x)$.

(b): On the one hand, suppose the axiom of substitution. Now consider some function $F$ whose domain $D$ is a set. Then $F''(D) = \mathrm{Ran}(F)$. Because $F''(D)$ is a set, then, because $D \times F''(D)$ is a set, and $F$ is a subclass of $D \times F''(D)$, $F$ is a set. On the other hand, suppose that any function whose domain is a set is also a set. Now consider any function $F$ and any set $x$. Then $F''(x)$ is a function whose domain is $x$, which is a set, and thus $F''(x)$ is also a set by assumption.

## Exercise 3.2

Suppose the axiom of choice, and that any class that can be put into a 1-1 correspondence with a set is a set. Now consider any function $F$ and any set $x$. Let $C$ be the choice function for $x$. For each $y \in F''(x)$, let $S(y)$ be the set of all $z \in x$ such that $F(z) = y$ (and $S(y)$ is a set, as it is a subset of $x$, which is a set by assumption). Then $C(S(y)) \in S(y)$, that is, $C(S(y))$ is some $z \in x$ such that $F(z) = y$. Then let $f$ be the set of all ordered pairs $\langle y, C(S(y))\rangle$ for $y \in F''(x)$. This is a 1-1 correspondence between $F''(x)$ and the set of all elements $z \in x$ such that $z = C(S(y))$ for some $y \in F''(x)$. Hence, because $F''(x)$ can be put into a 1-1 correspondence with a set, it is a set.

# Section 4

## Notes

Note that every well ordered set is also properly well ordered, because obviously every proper lower section of the well ordered set would be a subset of the set, and hence also a set.